PYTHON PROGRAMMING



- INTRODUCTION AND WHY PYTHON?
- SETTING UP PYTHON AND JUPYTER NOTEBOOKS
- COURSE LAYOUT AND EXPECTATIONS

David Topping: david.topping@manchester.ac.uk

PYTHON PROGRAMMING

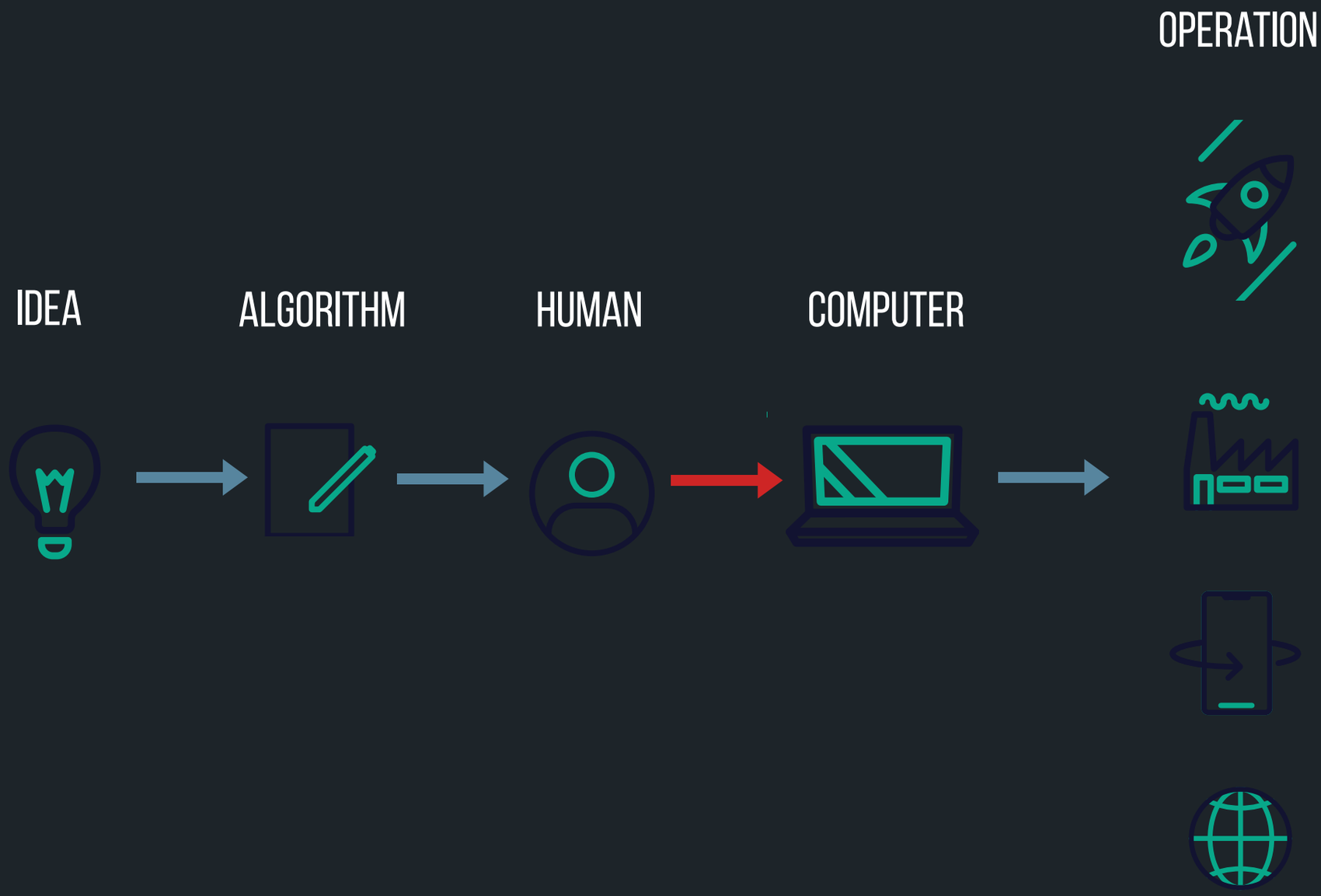


INTRODUCTION AND WHY PYTHON?

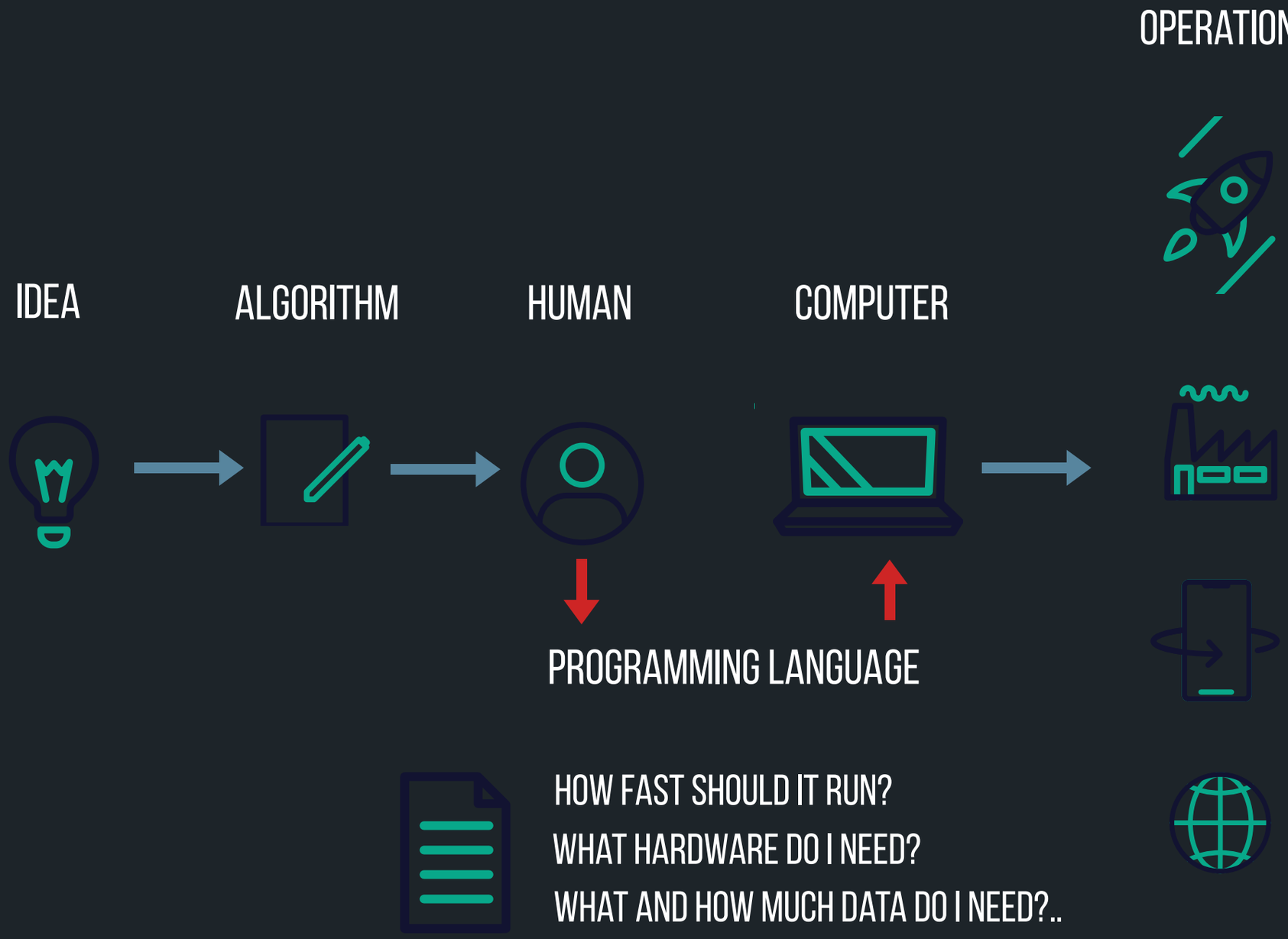
- WHY DO WE NEED PROGRAMMING LANGUAGES?
- A VERY VERY BRIEF HISTORY OF COMPUTING
- WHY PYTHON?

David Topping: david.topping@manchester.ac.uk

INTRODUCTION TO PROGRAMMING



INTRODUCTION TO PROGRAMMING



INTRODUCTION TO PROGRAMMING

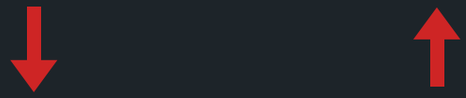
GRAB IMAGES
CAPTURE PERSONAL DATA
SELL DATA
MAKE MONEY





IDEA → ALGORITHM → HUMAN

COMPUTER → OPERATION



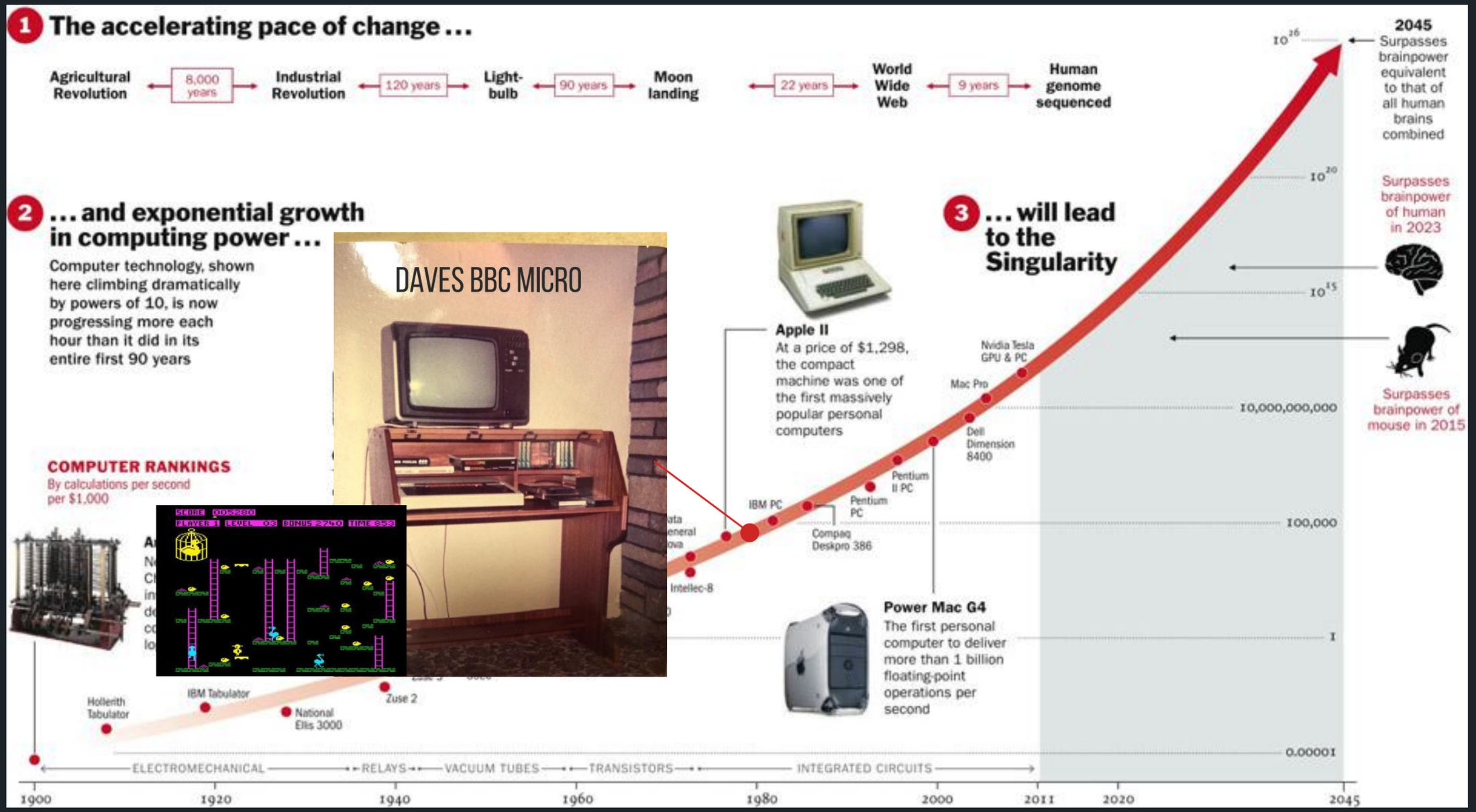
PROGRAMMING LANGUAGE ⟷ PYTHON: DJANGO FRAMEWORK

HOW FAST SHOULD IT RUN? ⟷ SHOULD BE RESPONSIVE!

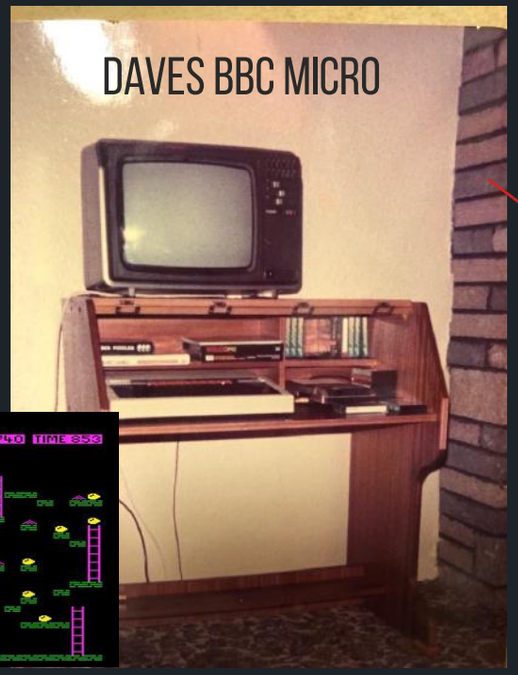
WHAT HARDWARE DO I NEED? ⟷ ANDROID, IOS - DATA CENTRES....

WHAT AND HOW MUCH DATA DO I NEED?.. ⟷ INDIVIDUAL DATA...

‘MODERN’ COMPUTING OFTEN DEFINES OUR CONVERSATIONS ABOUT THE TOPIC



‘MODERN’ COMPUTING OFTEN DEFINES OUR CONVERSATIONS ABOUT THE TOPIC





Several technological advancements over 70+ years led to the perfect AI storm: Algorithms + Data + Compute

Algorithms (1950s)
Emergence of the field of AI research around concept of machine being as intelligent as human being

Compute (1990s)
Breakthrough in semiconductor technology fuelling memory and compute capabilities (beginning of Moore’s law)

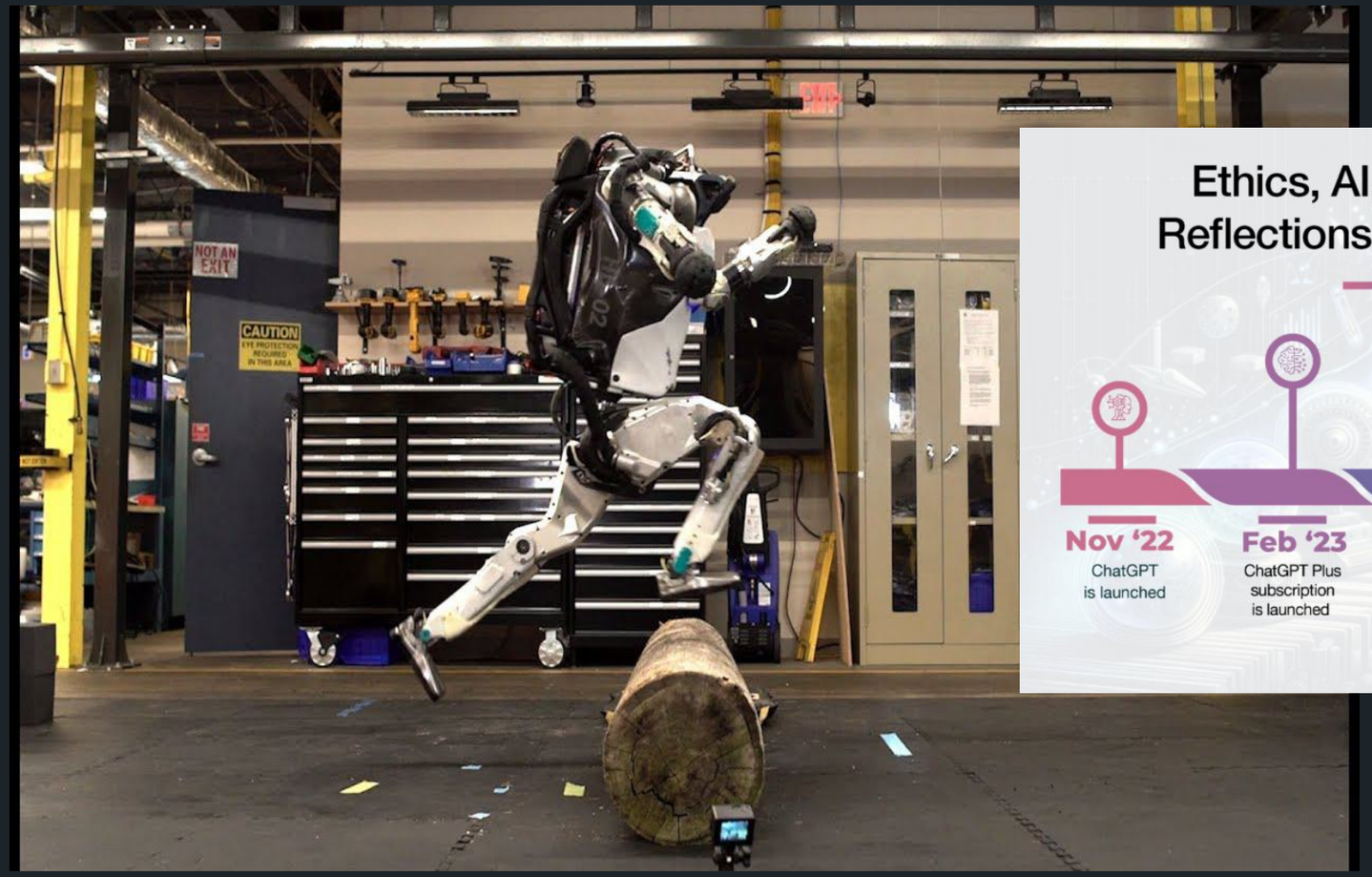
Data (2000s)
Large scale collection of data during Big Data hype; made actionable in datasets by cleaning, categorizing, and labelling

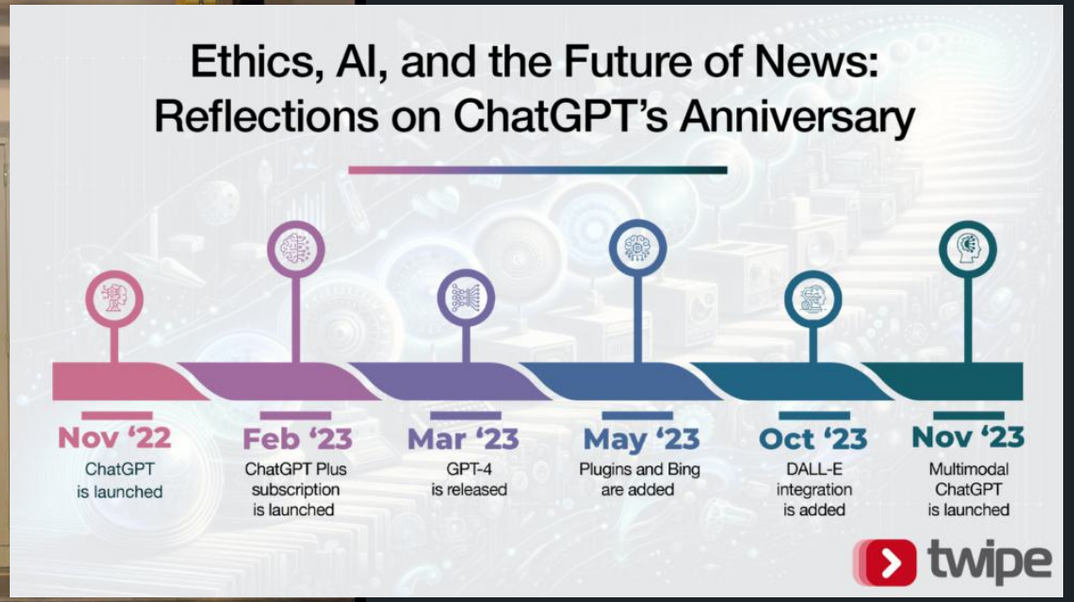
Attention (2017)
“Attention Is All You Need” paper proposing transformer models based on attention mechanism in NLP tasks

Chat (2022)
Making (generative) AI accessible to the masses through natural language-based chat interfaces

Source: Earlybird Analysis

‘MODERN’ COMPUTING OFTEN DEFINES OUR CONVERSATIONS ABOUT THE TOPIC

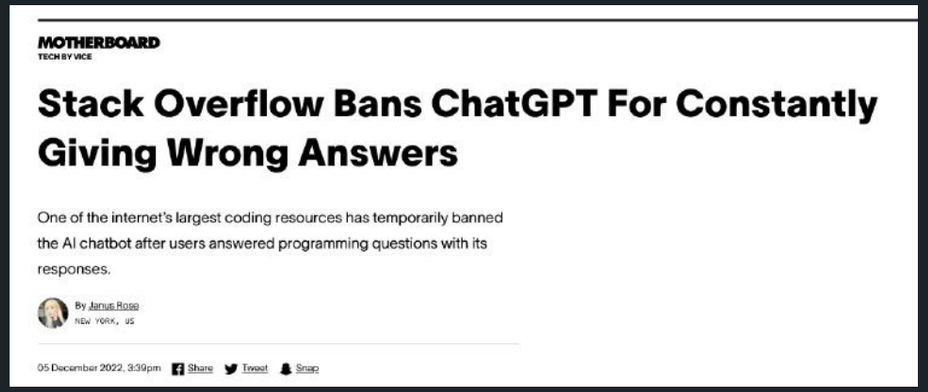




‘MODERN’ COMPUTING OFTEN DEFINES OUR CONVERSATIONS ABOUT THE TOPIC



2022



MOTHERBOARD
TECH BY VICE

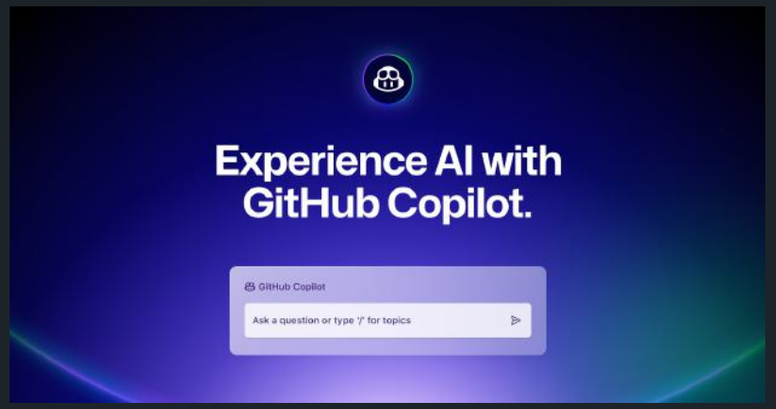
Stack Overflow Bans ChatGPT For Constantly Giving Wrong Answers

One of the internet's largest coding resources has temporarily banned the AI chatbot after users answered programming questions with its responses.

By Janus Rose
NEW YORK, US

05 December 2022, 3:39pm Share Tweet Snap

2023



COMPUTING HAS AN EXTENSIVE HISTORY



The Compendious Book on Calculation by Completion and Balancing (الكتاب المختصر في حساب الجبر والمقابلة)

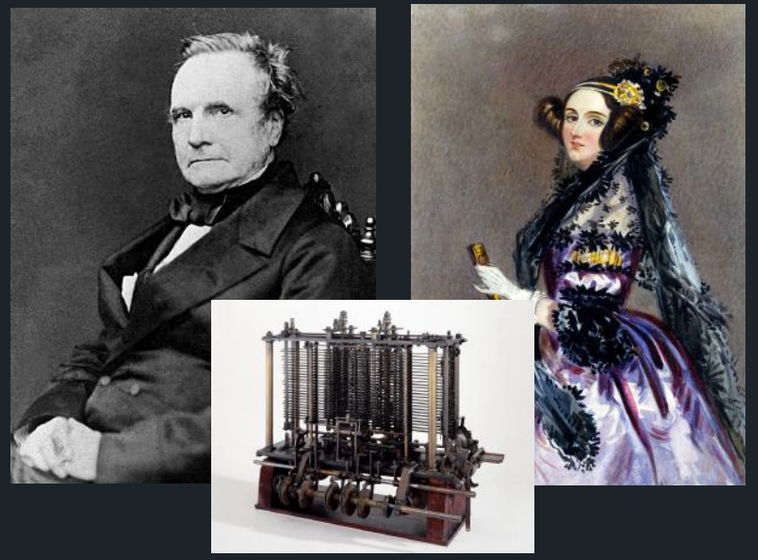
Al-Khwārizmī, in full Muḥammad ibn Mūsā al-Khwārizmī, (born c. 780—died c. 850)

MUSLIM MATHEMATICIAN AND ASTRONOMER WHOSE MAJOR WORKS INTRODUCED HINDU-ARABIC NUMERALS AND THE CONCEPTS OF ALGEBRA INTO EUROPEAN MATHEMATICS.

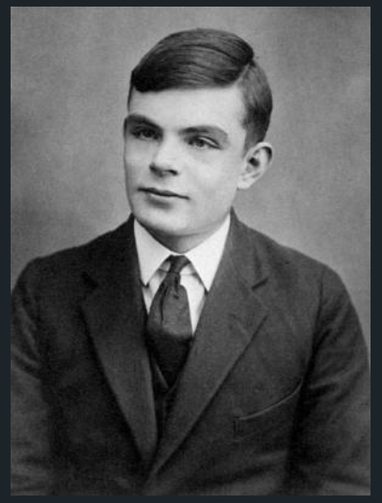
LATINIZED VERSIONS OF HIS NAME AND OF HIS MOST FAMOUS BOOK TITLE LIVE ON IN THE TERMS ALGORITHM AND ALGEBRA.

Charles Babbage, born 26 December 1791 London. The Hon. Augusta Ada Byron [Lovelace] Born 10 December 1815, London, England.

[THE ANALYTICAL ENGINE] MIGHT ACT UPON OTHER THINGS BESIDES NUMBER,...THE ENGINE MIGHT COMPOSE ELABORATE AND SCIENTIFIC PIECES OF MUSIC OF ANY DEGREE OF COMPLEXITY OR EXTENT.

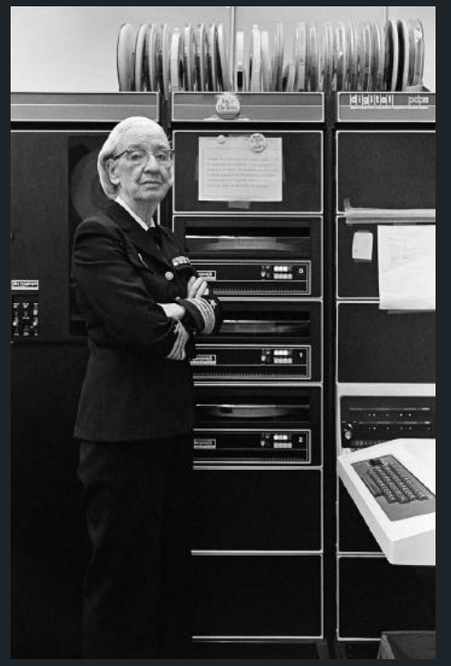


COMPUTING HAS AN EXTENSIVE HISTORY



Alan Mathison Turing (23 June 1912 – 7 June 1954) provided a formalisation of the concepts of algorithm and computation with the Turing machine, which can be considered a model of a general-purpose computer.

THE MANCHESTER BABY, ALSO KNOWN AS THE SMALL-SCALE EXPERIMENTAL MACHINE (SSEM), WAS THE WORLD'S FIRST ELECTRONIC STORED-PROGRAM COMPUTER. IT WAS BUILT AT THE UNIVERSITY OF MANCHESTER, UK, BY FREDERIC C. WILLIAMS, TOM KILBURN, AND GEOFF TOOTILL, AND RAN ITS FIRST PROGRAM ON 21 JUNE 1948



Grace Brewster Murray Hopper (December 9, 1906 – January 1, 1992) an American computer scientist and United States Navy rear admiral.

SHE POPULARIZED THE IDEA OF MACHINE-INDEPENDENT PROGRAMMING LANGUAGES, WHICH LED TO THE DEVELOPMENT OF COBOL, AN EARLY HIGH-LEVEL PROGRAMMING LANGUAGE STILL IN USE TODAY.

THE COMPUTER...WAS ORIGINALLY A PERSON!



In 1852 Radhanath Sickdhar managed to find out the height of Peak XV, an icy peak in the Himalayas – Mount Everest.
SPENT 4 YEARS UNSCRAMBLING MATHEMATICAL DATA FROM SEVERAL THOUSAND INDIANS AND WAS NAMED THE GREAT TRIGONOMETRICAL SURVEY (GTS) IN 1819. WAS CALLED A "COMPUTER" SINCE HE WORKED ON COMPUTATION OF DATA AND WAS PROMOTED TO THE POSITION OF "*CHIEF COMPUTER*" BECAUSE OF HIS GOOD WORK.

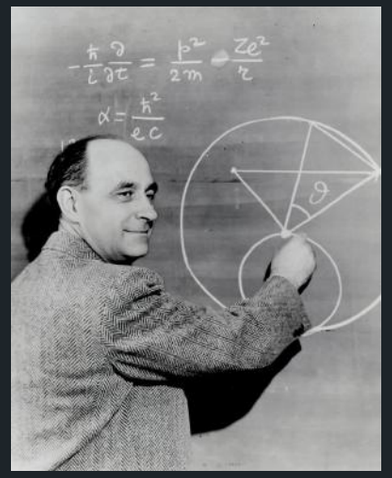
the "West Area Computers" at NASA [see Hidden figures].
Katherine Coleman Goble Johnson
"*While the specific tasks a computer did varied according to need and her department, the majority of computing work involved three components: reading film, running calculations, and plotting data.*"





PROGRAMMING LANGUAGES HAVE DEVELOPED AT A RAPID PACE...AS HAS THE ROLE OF SCIENTISTS

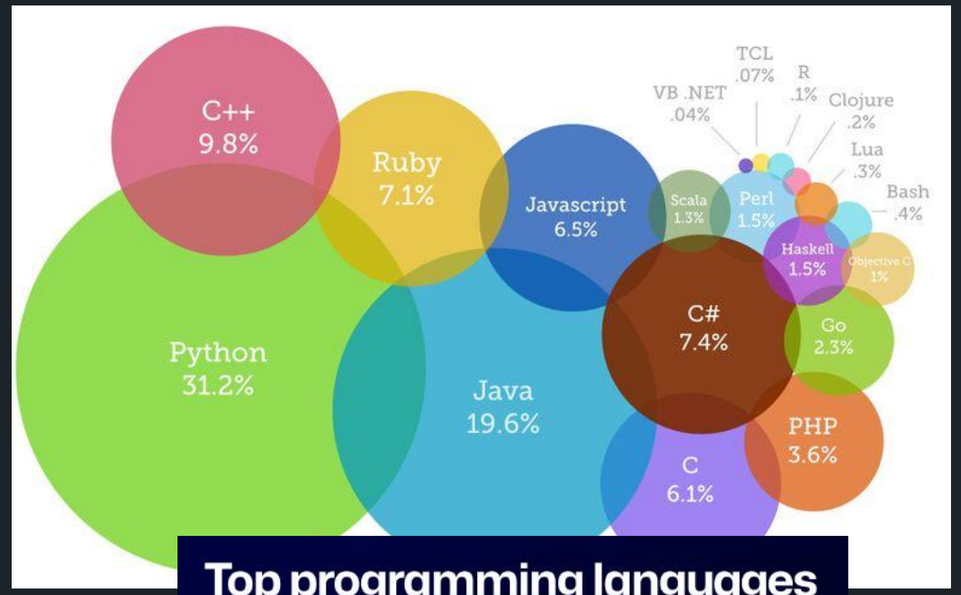
SMALL NUMBER OF APPLICATIONS AND DOMAIN EXPERTS



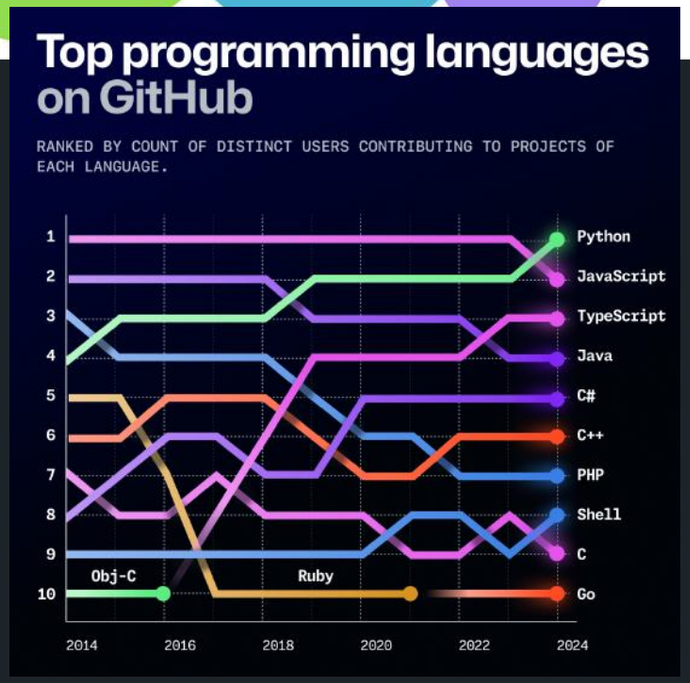
LARGE NUMBER OF APPLICATIONS AND DOMAIN EXPERTS



WHY PYTHON?



TOP 10 PROGRAMMING LANGUAGES 2020
HTTPS://WWW.DEVSARAN.COM



HTTPS://STACKOVERFLOW.BLOG/2023/01/26/COMPARING-TAG-TRENDS-WITH-OUR-MOST-LOVED-PROGRAMMING-LANGUAGES/

C/C#/C++/OBJECTIVE C:
THE OLDEST - 'LOW-LEVEL' LIBRARIES

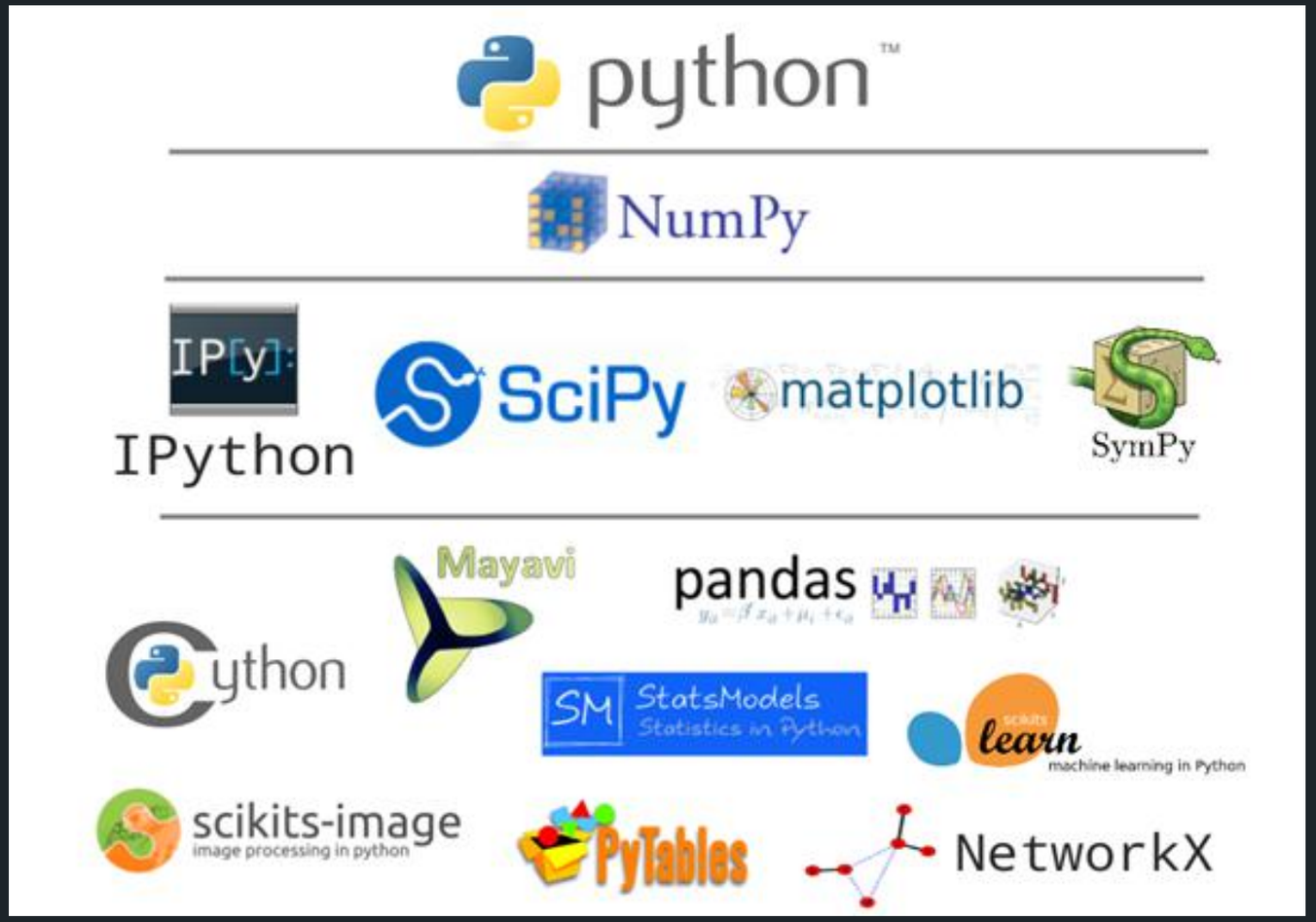
PERL:
WEB/DESKTOP APPS

JAVA/PHP:
WEB DEVELOPMENT

...WHERE IS FORTRAN??

PYTHON:
WEB APPS, DATA ANALYSIS, USER INTERFACES, AND MUCH MORE...

WHY PYTHON?



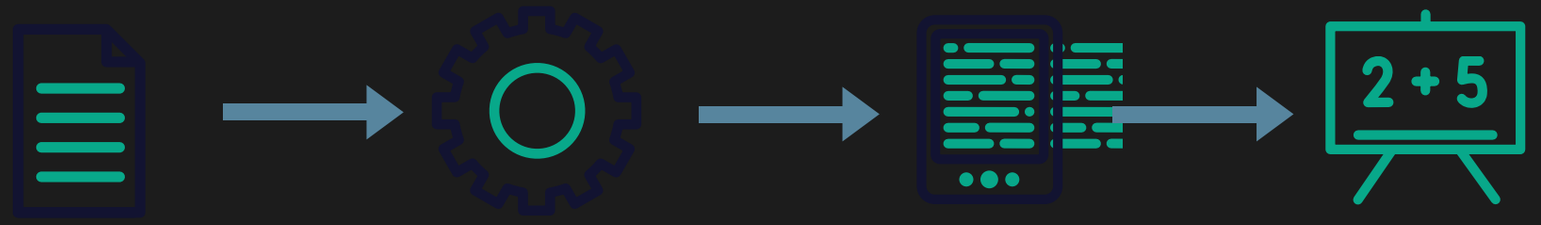
- LOTS OF EXISTING APPLICATIONS
- NUMERICAL MODELLING TO DATA ANALYSIS
- HUGE ONLINE COMMUNITY, AND SUPPORT
- TRANSFERABLE SKILLS

PROGRAMMING LANGUAGES

WHAT WE WRITE → WHAT WE WANT

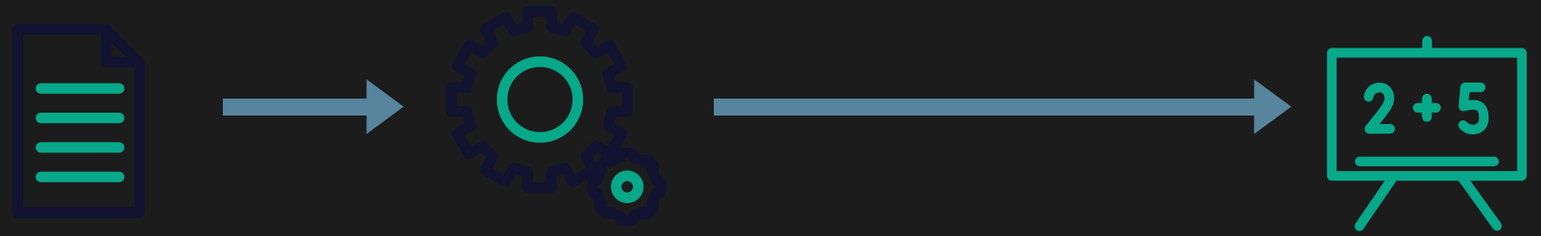
COMPILED LANGUAGE
[C++/FORTRAN..]

SOURCE CODE → COMPILER → MACHINE CODE → OUTPUT



INTERPRETED LANGUAGE
[JAVASCRIPT, PYTHON..]

SOURCE CODE → INTERPRETER → OUTPUT

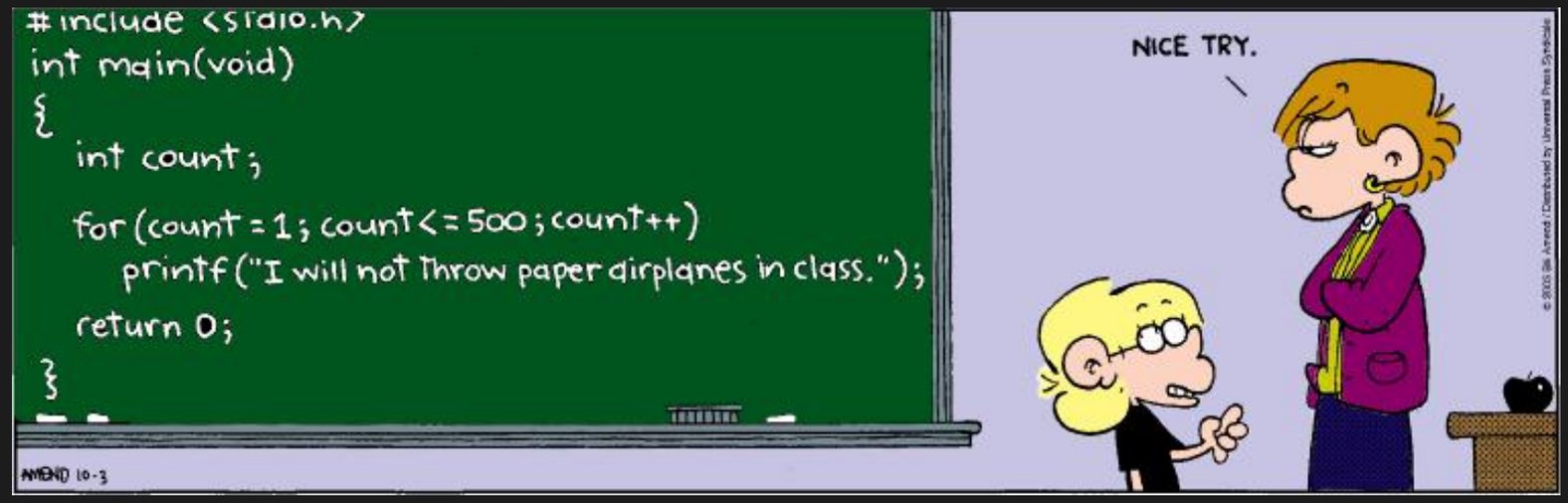


PROGRAMMING LANGUAGES

WHAT WE WRITE → WHAT WE WANT

THE SYNTAX OF A PROGRAMMING LANGUAGE IS THE SET OF RULES THAT CREATE STRUCTURALLY ALLOWED PROGRAMS





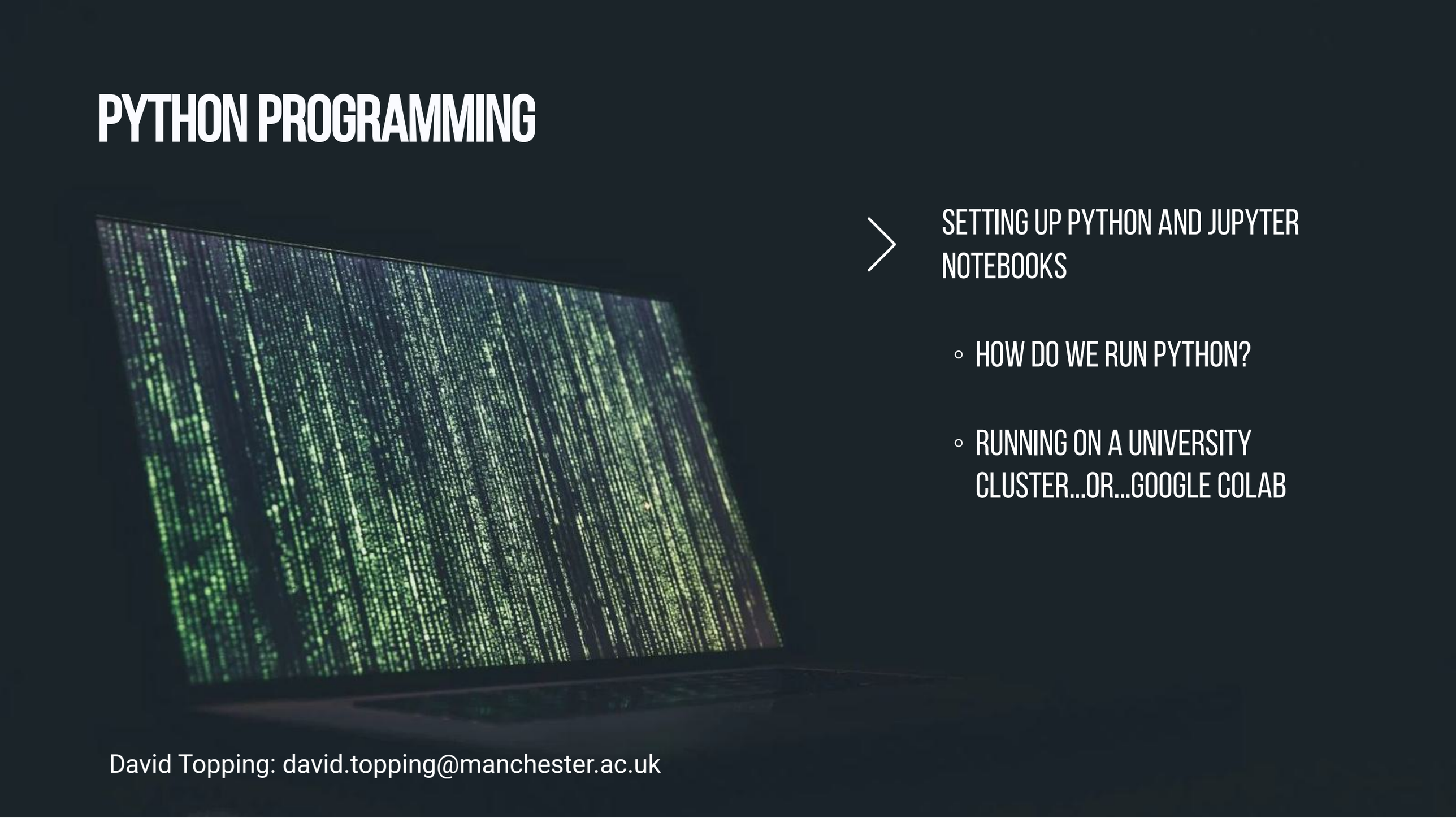
PYTHON PROGRAMMING



- INTRODUCTION TO PROGRAMMING: WHY PYTHON?
- SETTING UP PYTHON AND JUPYTER NOTEBOOKS
- COURSE LAYOUT

David Topping: david.topping@manchester.ac.uk

PYTHON PROGRAMMING



SETTING UP PYTHON AND JUPYTER NOTEBOOKS

- HOW DO WE RUN PYTHON?
- RUNNING ON A UNIVERSITY CLUSTER...OR...GOOGLE COLAB

David Topping: david.topping@manchester.ac.uk

RUNNING PYTHON

PERSONAL COMPUTER

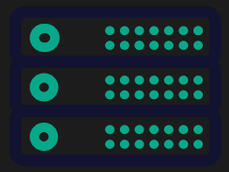


INTERACTIVE SHELL



COMMAND PROMPT
TERMINAL

SERVER



COMMAND LINE INTERPRETER + SOURCE CODE



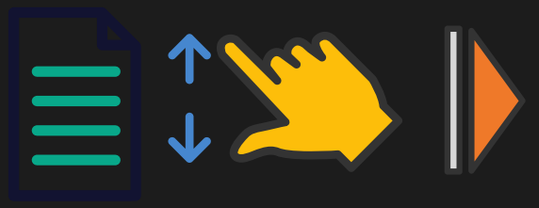
COMMAND PROMPT
TERMINAL



WEB SERVICE



JUPYTER NOTEBOOK VIRTUAL LAB BOOK + SOURCE CODE

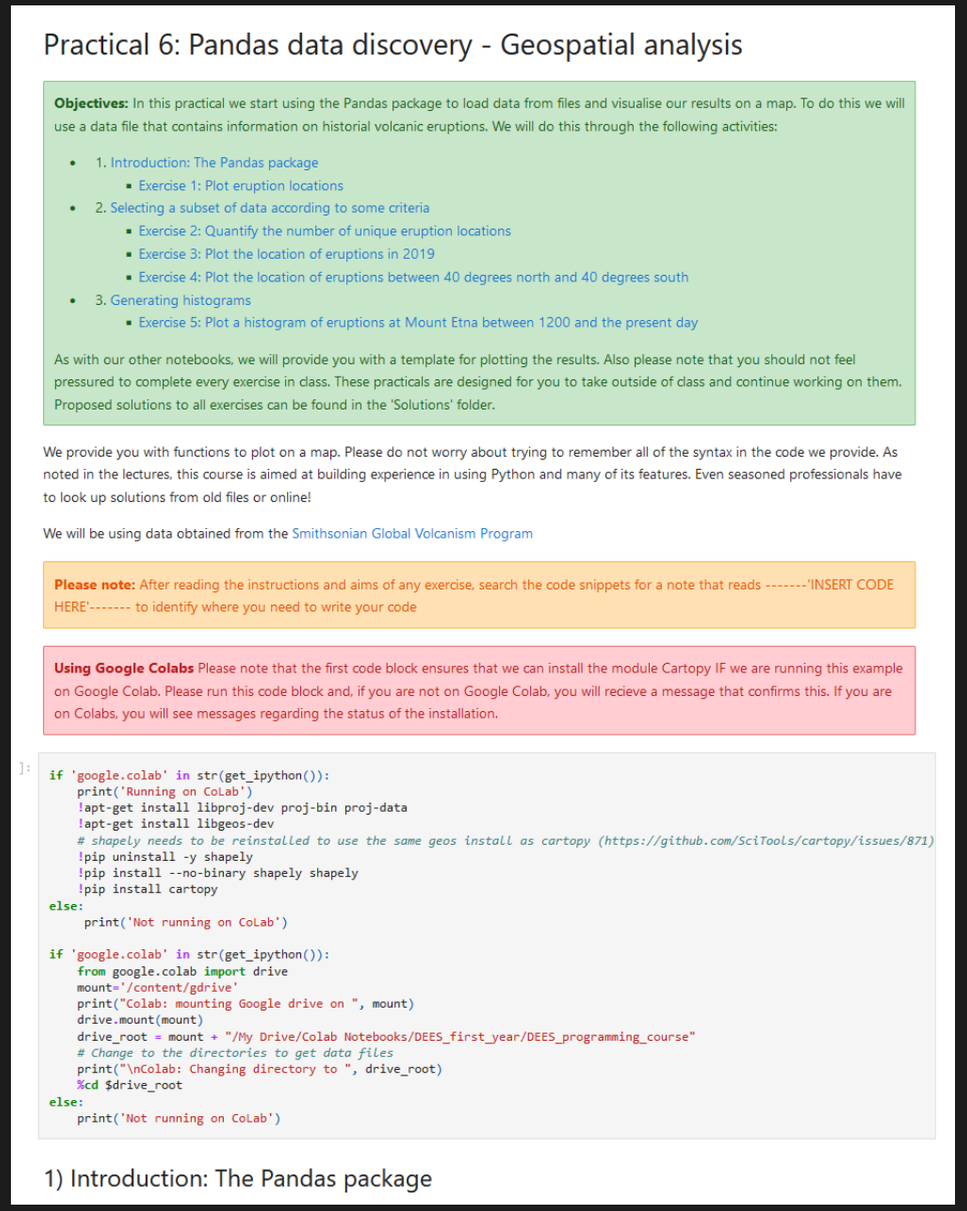


JUPYTER NOTEBOOKS

A nonprofit organization created to "develop open-source software, open-standards, and services for interactive computing across dozens of programming languages"

Jupyter's name is a reference to the three core programming languages supported by Jupyter, which are **Julia, Python and R**, and also a homage to Galileo's notebooks recording the discovery of the moons of Jupiter

It is like having a lab notebook where you can combine notes with running the experiment!



Practical 6: Pandas data discovery - Geospatial analysis

Objectives: In this practical we start using the Pandas package to load data from files and visualise our results on a map. To do this we will use a data file that contains information on historial volcanic eruptions. We will do this through the following activities:

- 1. Introduction: The Pandas package
  - Exercise 1: Plot eruption locations
- 2. Selecting a subset of data according to some criteria
  - Exercise 2: Quantify the number of unique eruption locations
  - Exercise 3: Plot the location of eruptions in 2019
  - Exercise 4: Plot the location of eruptions between 40 degrees north and 40 degrees south
- 3. Generating histograms
  - Exercise 5: Plot a histogram of eruptions at Mount Etna between 1200 and the present day

As with our other notebooks, we will provide you with a template for plotting the results. Also please note that you should not feel pressured to complete every exercise in class. These practicals are designed for you to take outside of class and continue working on them. Proposed solutions to all exercises can be found in the 'Solutions' folder.

We provide you with functions to plot on a map. Please do not worry about trying to remember all of the syntax in the code we provide. As noted in the lectures, this course is aimed at building experience in using Python and many of its features. Even seasoned professionals have to look up solutions from old files or online!

We will be using data obtained from the Smithsonian Global Volcanism Program

Please note: After reading the instructions and aims of any exercise, search the code snippets for a note that reads -------'INSERT CODE HERE'------- to identify where you need to write your code

Using Google Colabs Please note that the first code block ensures that we can install the module Cartopy IF we are running this example on Google Colab. Please run this code block and, if you are not on Google Colab, you will recieve a message that confirms this. If you are on Colabs, you will see messages regarding the status of the installation.

```
if 'google.colab' in str(get_ipython()):
    print('Running on CoLab')
    !apt-get install libproj-dev proj-bin proj-data
    !apt-get install libgeos-dev
    # shapely needs to be reinstalled to use the same geos install as cartopy (https://github.com/SciTools/cartopy/issues/871)
    !pip uninstall -y shapely
    !pip install --no-binary shapely shapely
    !pip install cartopy
else:
     print('Not running on CoLab')

if 'google.colab' in str(get_ipython()):
    from google.colab import drive
    mount='/content/gdrive'
    print("Colab: mounting Google drive on ", mount)
    drive.mount(mount)
    drive_root = mount + "/My Drive/Colab Notebooks/DEES_first_year/DEES_programming_course"
    # Change to the directories to get data files
    print("\nColab: Changing directory to ", drive_root)
    %cd $drive_root
else:
    print('Not running on CoLab')
```

1) Introduction: The Pandas package

ANACONDA PYTHON

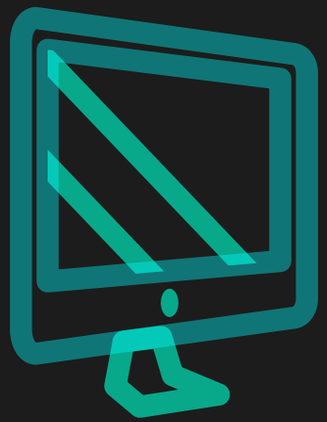
THE ANACONDA DISTRIBUTION : HTTPS://WWW.ANACONDA.COM/PRODUCTS/INDIVIDUAL
ACCEPT THE DEFAULT INSTALLATION OPTIONS.



SETTING UP A UNIVERSITY CLUSTER MACHINE

INSTRUCTIONS PROVIDED IN DOCUMENT ON BLACKBOARD
FIRST TIME WE LOG ON....

1) CREATE ENVIRONMENT



ANACONDA

2) GET CLASS CONTENT

GitHub

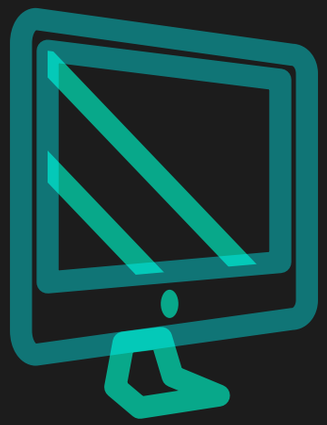
OneDrive

3) LINK AND STORE CLASS CONTENT

SETTING UP A UNIVERSITY CLUSTER MACHINE

INSTRUCTIONS PROVIDED IN DOCUMENT ON BLACKBOARD
SECOND TIME WE LOG ON....

1) LOAD ENVIRONMENT



ANACONDA®

2) GET CLASS CONTENT

OneDrive

SETTING UP ON YOUR GOOGLE ACCOUNT

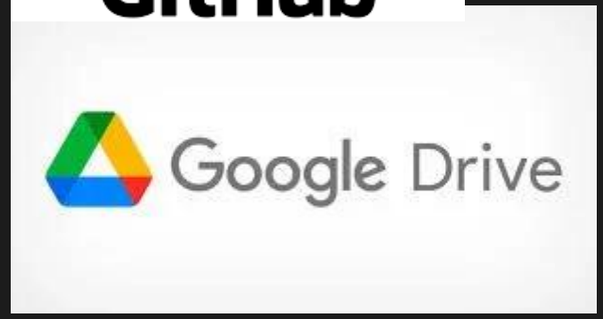
INSTRUCTIONS PROVIDED IN DOCUMENT ON BLACKBOARD
FIRST TIME WE LOG ON....

1) ACCESS GOOGLE COLAB



2) GET CLASS CONTENT





3) LINK AND STORE CLASS CONTENT

SETTING UP ON YOUR GOOGLE ACCOUNT

INSTRUCTIONS PROVIDED IN DOCUMENT ON BLACKBOARD
SECOND TIME WE LOG ON....

1) ACCESS GOOGLE COLAB



2) GET CLASS CONTENT



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'Vanilla' Python

Data types
Basic operations

Loops
Conditionals

Pandas and data discovery

Geospatial analysis

Statistical Analysis

Cluster Analysis

1 2 3 4 5 6 7 8 9

The Numpy module

Conway's Game Of Life

Differential Equations

Modules and Functions

Creating a notebook assessment
select topic from weeks 4 - 9

10 11 12

COURSE LAYOUT >>>>

LECTURE + PRACTICAL. SOLUTIONS ARE GIVEN

COURSE EXPECTATIONS

- GIVE YOURSELF TIME
- IT IS OK TO FIND THIS DIFFICULT
- NO ONE PRODUCES PERFECT CODE
- PRACTICE...PRACTICE...PRACTICE...

- TRY
- HELPING EACH OTHER IS FINE
- YOU ARE NOT EXPECTED TO REMEMBER SYNTAX PERFECTLY...
- CHATGPT IS FINE..I USE IT!
- PROGRAMMING IS A CONTINUAL JOURNEY...INTO A WONDEROUS TOMMOROW...



spock was ok in the end...

